SUBHASHIS SUARA

subhashissuara@gmail.com

subhashissuara.com ◇ linkedin.com/subhashissuara ◇ scholar.google.com/subhashissuara

EXPERIENCE

Site Reliability Engineer | European Organization for Nuclear Research (CERN) Sep 2023 – Aug 2026
Products: *GitLab, TWiki, Matomo, Confluence, Jira, Open Journal Systems, Discourse* *Geneva, Switzerland*
Technologies: *Kubernetes, Helm, Terraform, ArgoCD, CI/CD, Prometheus, Grafana, Fluent Bit, OpenSearch, Python, Go, Puppet, Perl, OpenShift, PHP, Java, Ruby, PostgreSQL, CephFS, S3, MySQL*

- Core member of the **GitLab** team, operated and improved CERN's GitLab infrastructure serving **18,000+ users**, **200,000+ projects**, and processing **50,000+ CI jobs daily**.
- Co-led **GitLab Geo** disaster recovery testing and storage failover exercises, validating business continuity and disaster recovery procedures in order to improve platform resilience.
- Contributed to migrating CERN's **GitLab runner infrastructure** from legacy systems to Kubernetes clusters, enabling faster scaling, improved operational efficiency, and greater flexibility to meet evolving user needs.
- Rolled out **GitLab security scanning** across the organization, providing development teams with automated security checks as part of their CI/CD workflows.
- Served as the service manager for **TWiki** (scientific documentation) and **Matomo** (web analytics), and backup service manager for **Confluence** & **Jira**, overseeing reliability, upgrades, lifecycle management, and user support.
- Investigated and resolved production incidents through debugging and root-cause analysis, producing incident reports, documenting corrective actions, and collaborating with users and teams to improve platform reliability and usability.

Openlab Summer Student | European Organization for Nuclear Research (CERN) Jun 2022 – Aug 2022
ReactJS, PHP, Docker, Kubernetes, Kibana, GitLab CI, VS Code Server, Drupal *Geneva, Switzerland*

- Developed a web platform enabling the CERN community to access logs and metrics for **1200+ Drupal websites** through the CERN Web Services Portal. [*Read the associated publication*]
- Built an on-demand remote development environment using **VS Code Server**, automating project setup and dependency management to improve developer productivity.
- Participated in technical lectures, workshops, and activities as part of the CERN Openlab Summer Student Programme.

Full Stack Developer Intern | Intrinsic Science Labs Oct 2021 – Feb 2022
ReactJS, NodeJS, ExpressJS, MySQL, Docker *Hyderabad, India – Remote*

- Designed and developed an online examination platform with a secure and intuitive user interface for candidates.
- Integrated **BigBlueButton** into the platform to enable real-time virtual classrooms and teacher–student collaboration.
- Built an exam question creation tool with **LaTeX** support, simplifying mathematical equation input and improving question creation efficiency.

Backend Developer Intern | Knowledge Lens May 2021 – Jul 2021
Flask, ArangoDB, MinIO *Bengaluru, India – Remote*

- Developed backend services and request handlers for an internal intranet portal and dashboard reporting system.
- Implemented automation scripts for data backups, database operations, and data extraction into structured JSON formats.

EDUCATION

XIM University Bhubaneswar, Odisha, India
Bachelor of Technology (Honors) in Computer Science & Engineering *Jun 2019 - Apr 2023*
CGPA: 9.54/10
Vice Chancellor's Medal for Academic Excellence
Bachelor's Thesis: Explainable Deep Learning in Medical Imaging with Grad-CAM

PROJECTS

Abuse and Anomaly Detection using Machine Learning for GitLab Runners: Mentored a student in developing a prototype to detect potential misuse of CERN's GitLab Runner infrastructure using Prometheus CPU and memory metrics, Isolation Forest, and an LSTM autoencoder. The project explored ML-based anomaly detection to identify unusual workload patterns and improve resource monitoring in shared CI/CD environments. [*Read the associated publication*]

Smart Health Monitoring System: A real-time health monitoring system built using ReactJS, NodeJS, ExpressJS, and MongoDB, featuring a custom dashboard and integrating an IoT device (Raspberry Pi) to collect and stream vital patient data such as heart rate, oxygen saturation, and body temperature to a backend server via WebSockets, enabling continuous monitoring and timely clinical insights.

MyCampus: A campus-focused social networking app that connects users based on their personality, interests, and lifestyle. It features a user-friendly authentication flow, a comprehensive questionnaire for profile creation, a matching engine, and a secure chat system built using Flutter, Flask, and Firebase, enabling users to build meaningful connections.

QubeBot: A homegrown WhatsApp chatbot developed using Python and Selenium WebDriver, built before the availability of WhatsApp Business chat solutions. It delivers information such as weather updates, top news headlines, instant search results, and song lyrics via messages, and can be used for FAQ-style interactions for user queries.

Above projects and more at: subhashissuara.com/#projects and **github.com/subhashissuara**

SKILLS

Infrastructure	Kubernetes, OpenShift, Docker, Helm, Terraform, ArgoCD, Puppet, GitLab Runner
Observability	Prometheus, Grafana, Fluent Bit, OpenSearch, Kibana
Languages	Python, Go, PHP, JavaScript, Ruby, Perl, Java, C/C++
Databases	PostgreSQL, MySQL, CephFS, S3, MongoDB
Tools	Linux, Git, Bash, GitLab, GraphQL, Postman, LaTeX

PUBLICATIONS

I. P. Trobo, K. Evangelou, **S. Suara**. *Running Version Control Systems and Continuous Integration to Fulfill CERN and LHC Experiments Needs.*
EPJ Web of Conferences, Vol. 337 (2025)
27th International Conference on Computing in High Energy and Nuclear Physics (CHEP 2024)
DOI: 10.1051/epjconf/202533701276

S. Suara, A. Jha, P. Sinha, A. A. Sekh. *Is Grad-CAM Explainable in Medical Images?*
Communications in Computer and Information Science (Springer)
International Conference on Computer Vision and Image Processing (CVIP 2023)
DOI: 10.1007/978-3-031-58181-6_11

COMMUNITY & PROFESSIONAL INVOLVEMENT

- Attended the **Thematic CERN School of Computing on Security (2025)**, organized by CERN in collaboration with **UKRI STFC**, with a focus on security in large-scale distributed and scientific computing systems.
- Attended **Web Summit 2024** in Lisbon, participating in industry and research talks on cloud computing, infrastructure, and emerging technologies.
- Volunteered as staff for **Kubernetes Community Days / Cloud Native Suisse Romande** at CERN, supporting community-driven events focused on cloud-native technologies and open-source ecosystems.
- Conducted guided tours at **CERN**, introducing visitors from diverse backgrounds to large-scale computing infrastructure and research facilities at CERN.
- **Winner** of Trithon Hackathon 2020 among 160 teams.
- President, **Xplore - The Computing Society**, XIM University *Dec 2019 – June 2022*